# Six clues in a slow fall

A Coffee Filter Parachute

Name _____ Date _____



Big held the coffee filter open like a tiny canopy. Gravity pulled the parachute toward the floor. As it moved, tiny moving pieces of air pushed back. This air resistance, or drag, slowed the fall but did not stop gravity.



Little dropped the filter flat, then crumpled the same filter and dropped it again. Its weight had not changed, but its shape had. The wide filter showed more surface area to the air, made more drag, and floated down longer.



Next they kept the canopy and strings the same. One parachute carried one paperclip; another carried two. The second paperclip added weight, so gravity pulled down harder while the same canopy provided drag. In this test, that parachute descended faster. Without air, the extra paperclip would not make it fall faster.

| | |
|---|---|
| **gravity** | the pull that draws objects toward Earth |
| **air** | tiny moving pieces of gas all around us |
| **air resistance / drag** | air pushing against an object moving through it |
| **canopy** | the wide top of a parachute |
| **surface area** | how much outside space a surface covers |
| **weight** | the pull of gravity on an object |
| **fair test** | a test that changes one thing at a time |
| **predict** | say what you think will happen before a test |
| **measure** | use numbers to describe what happens |

# Six clues in a slow fall

A Coffee Filter Parachute

Name _____ Date _____



Big suggested a fair test: change one thing at a time. They predicted a bigger canopy would fall more slowly, then kept the weight and string length equal. They measured several drops because one wobbly fall could fool them.



Sometimes the canopy rocked from side to side as air spilled unevenly around its rim. A small hole at the top let a little air pass through. The parachute wobbled less, so its path became steadier.



Their paper parachute was small, but its mechanism was real. Parachutes for people and space landers also spread a large canopy through air. Gravity keeps pulling down while drag pushes against motion, helping the load land more slowly.

1. Why does a flat coffee filter fall more slowly than the same filter crumpled?
2. How do repeated drops make the result of a fair test more trustworthy?
3. How do gravity and drag work together when a parachute descends?

## Your turn

Write one parachute prediction and name what you would measure.

Example: I predict the bigger canopy will fall longer, so I will measure its drop time.